

Structure vs Randomness: Determinism, Chaos, and Non-Uniqueness (TNA Core)

Abstract

Determinism in classical physics relies on the assumption that identical initial conditions yield a unique trajectory. This assumption is mathematically grounded in Lipschitz continuity. When this condition fails, systems may admit multiple valid evolutions from the same initial state. This paper formalizes the claim that what is often interpreted as “randomness” is instead the observable projection of structural non-uniqueness. Using the TNA framework, we distinguish between ontological multiplicity (non-Lipschitz dynamics) and epistemic unpredictability (non-computability), and propose that randomness emerges when structure fails to enforce a unique continuation.

1. Introduction

Classical determinism assumes a one-to-one mapping between initial conditions and trajectories. However, this assumption depends on regularity conditions that are often implicit.

We ask:

Is randomness intrinsic, or does it arise when structure fails to enforce uniqueness?

2. Lipschitz Domain: Structural Closure and Uniqueness

Consider:

$$[\dot{x} = f(x, t)]$$

If (f) is Lipschitz in (x) , then:

- A unique solution exists
- The system is deterministic in a strong sense

TNA Formalization

Let:

- (E) : observable state space
- $(\Omega \subset E)$: admissible domain
- (F_{min}) : minimal structural bound such that

$$[|f(x) - f(y)| \leq F_{min}|x - y|]$$

Then:

Lipschitz continuity is equivalent to the existence of a finite structural bound (F_{min}) that enforces uniqueness.

Interpretation:

- Structure (N_1) is sufficient
 - Evolution is closed
 - $(\Delta = \emptyset)$
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3. Breakdown of Lipschitz: Structural Multiplicity

If Lipschitz fails:

$$[\sup \frac{|f(x) - f(y)|}{|x - y|} = \infty]$$

Then:

- No finite (F_{min}) exists
- Structural closure fails
- Multiple continuations emerge

$$[\Delta \neq \emptyset]$$

This is **ontological multiplicity**, not probabilistic randomness.

4. Robust Physical Examples of Non-Uniqueness

4.1 Coulomb Friction

$$[m\ddot{x} = -\mu \operatorname{sign}(\dot{x})]$$

At ($\dot{x} = 0$):

- System may remain at rest or start moving
- Multiple valid trajectories exist

4.2 Filippov Systems (Discontinuous Dynamics)

Piecewise-defined vector fields generate:

- Sliding modes
- Set-valued dynamics

These are not pathological—they are common in:

- Mechanical contact
 - Control systems
 - Impact dynamics
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5. Chaos vs Structural Non-Uniqueness

Chaos:

- Unique trajectory
- Sensitive dependence

Non-Lipschitz:

- Multiple trajectories
- No unique evolution

Conclusion:

6. Two Sources of Apparent Randomness

6.1 Structural Multiplicity (Ontological)

- Multiple solutions exist
- No selection rule

6.2 Computational Inaccessibility (Epistemic)

- Unique solution exists
- Cannot be computed or predicted

These must not be conflated.

7. TNA–Lipschitz Correspondence: Theorem and Corollaries

7.1 Preliminaries

Consider the dynamical system:

$$\dot{x} = f(x), \quad x \in \Omega \subseteq \mathbb{R}^n$$

with f continuous.

Define:

- $\Delta(x_0)$: set of admissible trajectories from x_0
 - F_{min} : minimal Lipschitz bound ensuring uniqueness
 - $\partial\Omega$: region where Lipschitz regularity fails
 - F_B^{logic} : structural multiplicity (measure over Δ)
 - F_B^{dyn} : observable divergence of trajectories
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7.2 The TNA–Lipschitz Theorem (Directional Form)

Theorem (TNA–Lipschitz Correspondence, rigorous form).

Let f be locally Lipschitz on Ω . Then:

$$\exists F_{min} < \infty \text{ such that } \|f(x) - f(y)\| \leq F_{min} \|x - y\|$$

implies:

$$|\Delta(x_0)| = 1 \quad \forall x_0 \in \Omega$$

(i.e., uniqueness of trajectories).

Reciprocal (TNA interpretation, partial)

If Lipschitz regularity fails at x_0 , then at least one of the following holds:

- non-uniqueness: $|\Delta(x_0)| > 1$
- or failure of well-defined evolution

Thus:

Loss of Lipschitz regularity is a necessary condition for structural multiplicity.

7.3 Minimality Condition (TNA refinement)

Define F_{min} as the infimum over all admissible Lipschitz constants ensuring uniqueness.

Then:

$$\forall F < F_{min}, \quad \exists x_0 \in \Omega \text{ such that } |\Delta(x_0)| \neq 1$$

This encodes:

F_{min} as the minimal structural constraint required for coherent evolution.

7.4 Boundary Condition

If:

$$F_{min} \rightarrow \infty$$

then:

$$x_0 \in \partial\Omega \Rightarrow |\Delta(x_0)| > 1 \text{ or undefined}$$

This defines the transition:

- interior $\rightarrow$ structured evolution
 - boundary $\rightarrow$ multiplicity / breakdown
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7.5 Corollaries

Corollary 1 — Structural Multiplicity

If no finite F_{min} exists locally, then:

$$|\Delta(x_0)| > 1$$

Corollary 2 — Emergent Randomness

If:

$$|\Delta(x_0)| > 1$$

and no selection rule is specified, then observed randomness arises as:

■ *the macroscopic manifestation of unresolved multiplicity.*

Corollary 3 — Epistemic vs Ontological Regimes

- $F_{min} < \infty$, but computationally inaccessible $\rightarrow$ **epistemic randomness**
 - $F_{min} \rightarrow \infty \rightarrow$ **ontological multiplicity**
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Corollary 4 – Structural vs Dynamical Pressure

Let:

- $F_B^{logic} = \mu(\Delta)$
- $F_B^{dyn} \sim \sup \|x_1(t) - x_2(t)\|$

Then:

F_B^{dyn} can be interpreted as a **projection** of F_B^{logic} onto observable trajectory space.

(Conceptual mapping; no specific functional form assumed.)

Corollary 5 – Regularization as Selection Mechanism

Any mechanism enforcing:

$$F_{min} < \infty$$

acts as a **selection operator**:

$$\Delta \rightarrow \{\text{single trajectory}\}$$

Examples:

- Filippov selection rules
- friction regularization
- quantum state reduction (*mechanism hypothetical*)

7.6 TNA Correspondence

DYNAMICAL SYSTEMS	TNA INTERPRETATION
Lipschitz bound	F_{min}
Unique trajectory	Structural closure
Non-Lipschitz	$\partial\Omega$
Multiple solutions	$\Delta \neq \emptyset$
Divergence	F_B^{dyn}

DYNAMICAL SYSTEMS	TNA INTERPRETATION
Multiplicity	F_B^{logic}

7.7 Conceptual Synthesis

$$\text{Structure} \longleftrightarrow F_{min} \longleftrightarrow \text{Uniqueness}$$

7.8 Link to TNA Core Principle

This establishes the dynamical analogue of TNA:

■ *A system requires a minimal structural constraint F_{min} to sustain coherent evolution.*

8 — Classification of Dynamical Regimes via TNA

8.1 Motivation

Classical dynamics distinguishes between deterministic and stochastic systems. However, this dichotomy is insufficient.

From the TNA perspective, system behavior is governed not by determinism vs randomness, but by the **structural conditions that enforce or fail to enforce uniqueness**.

8.2 Structural Classification Principle

We classify dynamical regimes according to the behavior of:

- F_{min} : minimal structural constraint
- $\Delta(x_0)$: set of admissible trajectories

8.3 Regime I — Structural Closure (Lipschitz Domain)

Condition:

$$F_{min} < \infty \quad \text{and finite}$$

Properties:

- Unique trajectory
- Stable evolution
- Predictability (in principle)

Interpretation:

▮ *Structure fully constrains evolution.*

Examples:

- Classical mechanics (smooth potentials)
 - Regular ODE systems
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8.4 Regime II — Deterministic Chaos

Condition:

$$F_{min} < \infty \quad \text{but sensitivity} \rightarrow \infty$$

Properties:

- Unique trajectory
- Exponential divergence of nearby states
- Practical unpredictability

Interpretation:

▮ *Structure enforces uniqueness, but amplifies uncertainty.*

Examples:

- Lorenz system
- Turbulent flows (in certain regimes)

8.5 Regime III — Structural Multiplicity (Non-Lipschitz Domain)

Condition:

$$F_{min} \rightarrow \infty$$

Properties:

- Multiple valid trajectories
- No intrinsic selection rule
- Breakdown of classical determinism

Interpretation:

Structure fails to enforce a unique continuation.

Examples:

- Peano-type systems
 - Coulomb friction at zero velocity
 - Filippov systems
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8.6 Regime IV — Selection-Dominated Systems

Condition:

$$|\Delta(x_0)| > 1 \quad + \quad \text{external or internal selection mechanism}$$

Properties:

- Multiplicity exists at the structural level
- One trajectory is realized
- Apparent randomness emerges

Interpretation:

Observed evolution is the result of a selection process over a multiplicity set.

Examples:

- Filippov selection rules
- Regularized friction
- Quantum state reduction (hypothetical)

8.7 Regime V – Epistemic Non-Computability

Condition:

$F_{min} < \infty$ but not computationally accessible

Properties:

- Unique trajectory exists
- Cannot be computed or predicted
- Appears random

Interpretation:

Randomness arises from computational limits, not structural multiplicity.

Examples:

- Systems with uncomputable dynamics (Penrose hypothesis)
- Extreme chaotic systems beyond numerical resolution

8.8 Unified Table

REGIME	F_{min}	$ \Delta $	BEHAVIOR	INTERPRETATION
I – Closure	finite	1	Deterministic	Full structure
II – Chaos	finite	1	Unpredictable	Sensitive structure
III – Multiplicity	∞	>1	Non-unique	Structural failure
IV – Selection	$\infty + \text{rule}$	1 observed	Random-like	Selection over multiplicity
V – Epistemic	finite (inaccessible)	1	Random-like	Computational limit

8.9 Conceptual Synthesis

All dynamical behavior can be expressed as:

$$\text{Observed Evolution} = \text{Structure} + (\text{Selection or Computation Limits})$$

8.10 Core TNA Statement

Determinism and randomness are not fundamental categories, but emergent regimes determined by structural constraints and their failure.

9. Penrose Hypothesis (Reinterpreted)

Penrose proposes gravitationally induced collapse of quantum states.

We reinterpret:

Gravity may act as a physical mechanism that reduces multiplicity (collapses (Δ)).

This is a structural hypothesis, not a formal equivalence.

10. Redefining Randomness

We define:

Randomness = the macroscopic experience of unresolved multiplicity.

Thus, randomness is not primitive, but emergent from structural conditions.

“Quantum randomness is not fundamental randomness but the macroscopic signature of unresolved structural multiplicity (Δ), requiring a physical selection mechanism.”

11. Implications

Physics

- Reinterpret stochasticity as structural underdetermination

Mathematics

- Lipschitz = structural control metric

Philosophy

- Determinism depends on structural completeness
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12. Conclusion

The classical dichotomy between determinism and randomness is insufficient.

A deeper structure emerges:

- Closed structure → uniqueness
- Open structure → multiplicity

Thus:

Randomness signals the failure of structure to enforce a single trajectory.

- **Filippov, A.F.** (1988). *Differential Equations with Discontinuous Righthand Sides*
 - **Penrose, R.** (1996). "On gravity's role in quantum state reduction"
 - **Kunze, M. & Monteiro Marques, M.D.P.** (2000).
 - **Arnold, V.I.** (1973). *Ordinary Differential Equations*
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Appendix A — Numerical Illustration of Non-Lipschitz Dynamics (Peano-Type System)

A.1 System Definition

Consider the scalar differential equation:

$$\dot{x} = \sqrt{|x|}$$

with initial condition:

$$x(0) = 0$$

This system is continuous but **not Lipschitz** at $x = 0$.

A.2 Analytical Solutions

This system admits **multiple valid solutions**:

Solution 1 (Trivial)

$$x(t) = 0$$

Solution 2 (Delayed Growth Family)

For any $t_0 \geq 0$:

$$x(t) = \begin{cases} 0 & t \leq t_0 \\ \frac{1}{4}(t - t_0)^2 & t > t_0 \end{cases}$$

Interpretation

- Same initial condition
- Infinite family of trajectories
- No internal rule selects one

Thus:

$$|\Delta(0)| = \infty$$

A.3 Structural Interpretation (TNA)

At $x = 0$:

- Lipschitz constant $F_{min} \rightarrow \infty$
- Structural constraint collapses
- System enters multiplicity regime

$$x_0 \in \partial\Omega$$

A.4 Numerical Simulation

A simple Euler discretization:

$$x_{n+1} = x_n + \Delta t \cdot \sqrt{|x_n|}$$

reveals:

- For $x_0 = 0$, the system **remains at 0** (numerical freezing)
 - Small perturbations ($x_0 = \epsilon$) produce divergence
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Observation

The numerical scheme implicitly enforces:

$$\Delta \rightarrow \text{single trajectory}$$

i.e., it acts as a **selection mechanism**.

A.5 Perturbation Experiment

Let:

$$x(0) = \epsilon, \quad \epsilon > 0$$

Then:

$$x(t) = \frac{1}{4}(t + C)^2$$

and trajectories diverge depending on ϵ .

Key Result

$$\lim_{\epsilon \rightarrow 0} x_{\epsilon}(t)$$

is **not unique**, confirming structural instability at the origin.

A.6 Connection to TNA–Lipschitz Framework

PROPERTY	OBSERVATION	TNA INTERPRETATION
Non-Lipschitz point	$x = 0$	Boundary $\partial\Omega$
Multiple solutions	Infinite family	$\Delta \neq \emptyset$
Numerical selection	Euler method	Artificial F_{min}
Sensitivity to perturbation	High	Structural multiplicity

A.7 Interpretation as “Emergent Randomness”

If measurement noise is present:

- System appears to "randomly choose" a trajectory
- But randomness is not intrinsic

It reflects:

A.8 Relevance to Physical Systems

This toy model captures structurally:

- Dry friction systems (Filippov dynamics)
 - Stick-slip transitions
 - Threshold activation processes
 - Candidate analogies for quantum state selection
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A.9 Conceptual Conclusion

This example demonstrates:

Non-Lipschitz $\Rightarrow \Delta \neq \emptyset \Rightarrow$ Selection required

Thus reinforcing the central claim:

Randomness emerges where structure fails to enforce uniqueness.